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Analysis of Salary Data

Data Scientist: Oliver März



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Agenda

Findings of linear regression modeling with tech salary data

- Data Description
- Regression Results
- Interpretation and Next Steps

Data Description

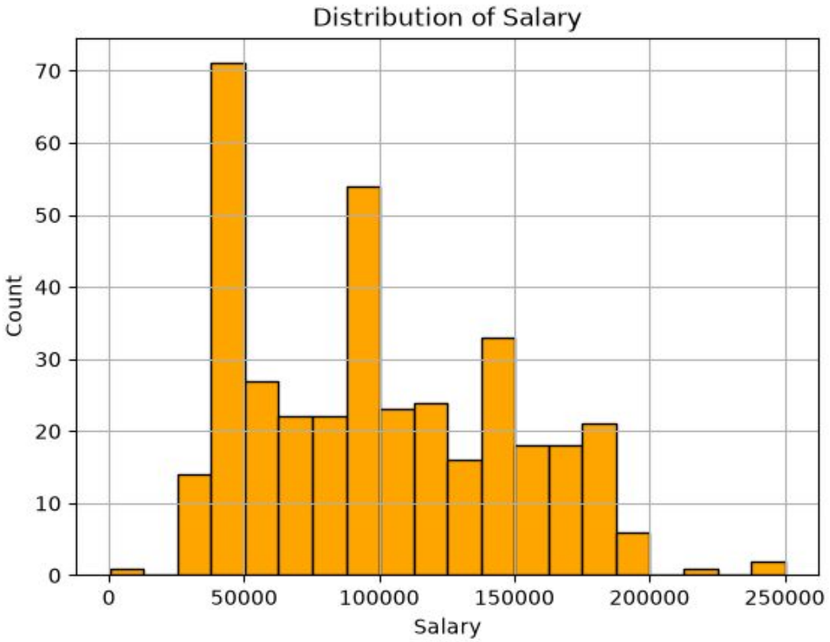
Employees

- 2 of the 375 rows had all values missing
- Those **2 rows were dropped**
- The number of employees in the dataset is: **373**

Data Description

- Range of the salaries: **350.00 to 250,000.00**
- The Male group earns **7.07 %** more than the Female group *
- The PhD group earns **111.14 %** more than the Bachelor's group *
- The PhD group earns **21.61 %** more than the Master's group *
- The Master's group earns **73.63 %** more than the Bachelor's group *
- More years of experience is associated with higher average salary (roughly linear upward trend)

* on average





Regression Results

Linear model predicting salary from years of experience, gender, job-title flags, and education (Bachelor's as baseline).

- R-squared = **0.915**
- Adjusted R-squared = **0.913**

OLS Regression Results

Dep. Variable:	salary	R-squared:	0.915
Model:	OLS	Adj. R-squared:	0.913
Method:	Least Squares	F-statistic:	391.7
Date:	Sun, 25 Oct 2026	Prob (F-statistic):	2.41e-187
Time:	15:18:31	Log-Likelihood:	-4090.6
No. Observations:	373	AIC:	8203.
Df Residuals:	362	BIC:	8246.
Df Model:	10		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
const	3.212e+04	2485.604	12.921	0.000	2.72e+04	3.7e+04
years_of_experience	4937.4278	197.548	24.994	0.000	4548.941	5325.914
is_male	7484.7304	1509.904	4.957	0.000	4515.446	1.05e+04
is_director	2.539e+04	3535.194	7.181	0.000	1.84e+04	3.23e+04
is_junior	-4492.5634	2458.516	-1.827	0.068	-9327.330	342.203
is_senior	1.307e+04	1976.833	6.610	0.000	9178.417	1.7e+04
is_manager	4434.5081	1960.352	2.262	0.024	579.400	8289.617
is_analyst	-176.3409	2103.878	-0.084	0.933	-4313.698	3961.016
is_engineer	2362.9515	3561.972	0.663	0.508	-4641.805	9367.708
has_master	1.345e+04	2053.896	6.547	0.000	9408.638	1.75e+04
has_phd	2.327e+04	2801.927	8.304	0.000	1.78e+04	2.88e+04

Omnibus:	130.784	Durbin-Watson:	1.815
Prob(Omnibus):	0.000	Jarque-Bera (JB):	1091.992
Skew:	1.230	Prob(JB):	7.53e-238
Kurtosis:	11.013	Cond. No.	63.1

Regression Results

Statistically significant vs. not significant variables

- **Significant** ($p < 0.05$):
  - years_of_experience
  - is_male
  - is_director
  - is_senior
  - is_manager
  - has_master
  - has_phd
- **Not significant**:
  - **is_analyst** and **is_engineer** are not significant, so we dropped them. They both have a p-value well above 0.05 with 0.933 and 0.508.
  - We also dropped **is_junior** with a p-value slightly above 0.05 with 0.068

Regression Results

2nd reduced model

- Kept variables:
  - years_of_experience
  - is_male
  - is_director
  - is_senior
  - is_manager
  - has_master
  - has_phd
- R-squared = 0.914
- Adjusted R-squared = 0.913

R-squared almost stayed the same after dropping the weak variables

OLS Regression Results

Dep. Variable:	salary	R-squared:	0.914
Model:	OLS	Adj. R-squared:	0.913
Method:	Least Squares	F-statistic:	557.4
Date:	Sun, 25 Oct 2026	Prob (F-statistic):	1.46e-190
Time:	17:04:05	Log-Likelihood:	-4092.6
No. Observations:	373	AIC:	8201.
Df Residuals:	365	BIC:	8233.
Df Model:	7		
Covariance Type:	nonrobust		

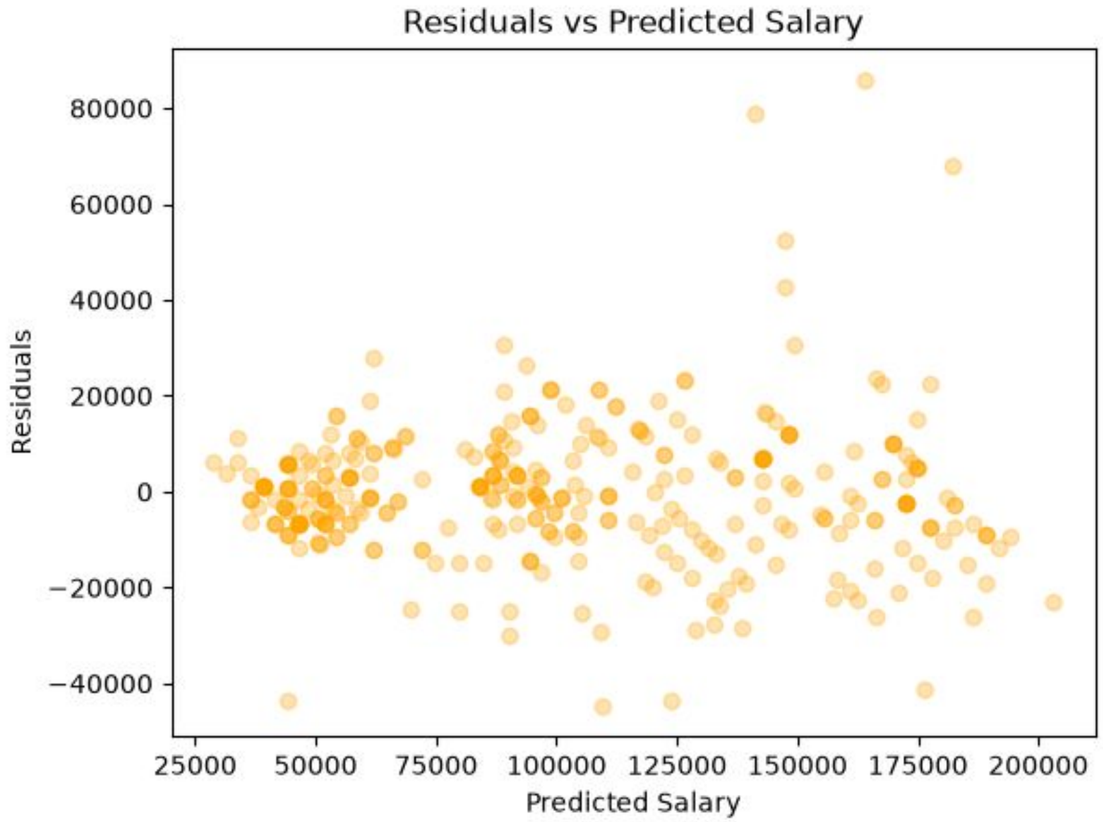
	coef	std err	t	P>|t|	[0.025	0.975]
const	2.876e+04	1677.411	17.146	0.000	2.55e+04	3.21e+04
years_of_experience	5104.4388	177.226	28.802	0.000	4755.927	5452.950
is_male	7629.6521	1481.834	5.149	0.000	4715.649	1.05e+04
is_director	2.498e+04	3503.028	7.131	0.000	1.81e+04	3.19e+04
is_senior	1.445e+04	1820.427	7.936	0.000	1.09e+04	1.8e+04
is_manager	4284.6595	1822.582	2.351	0.019	700.579	7868.740
has_master	1.404e+04	2032.078	6.908	0.000	1e+04	1.8e+04
has_phd	2.308e+04	2760.740	8.360	0.000	1.77e+04	2.85e+04

Omnibus:	126.635	Durbin-Watson:	1.790
Prob(Omnibus):	0.000	Jarque-Bera (JB):	1010.895
Skew:	1.196	Prob(JB):	3.07e-220
Kurtosis:	10.702	Cond. No.	60.8

Regression Results

How well the model fits (part 1)

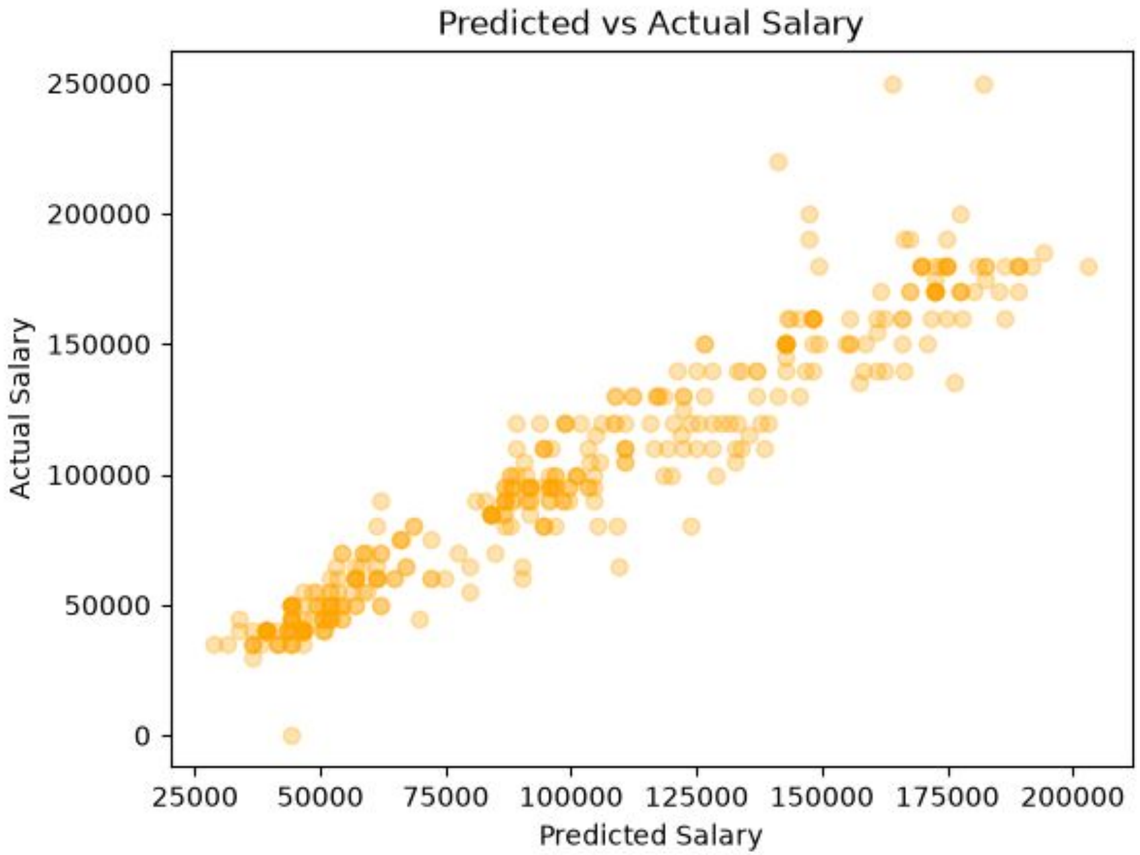
- The model fits well. R-squared = 0.914, so it **explains about 91% of the variation in salary**.
- Looking at the residuals-vs-predicted plot, **most residuals are scattered around 0** (roughly within +/- $20,000), which supports a decent linear fit. There are some larger outliers at higher predicted salaries, so it is not perfect.



Regression Results

How well the model fits (part 2)

- In the predicted vs. actual plot, most points follow the diagonal (where predicted ~ actual) pretty closely, with more spread at higher salaries.



Interpretation and Next Steps

Years of Experience `years_of_experience`

- With each year of experience the salary is on average **$5,104.44 higher** (with the other features fixed).
- We can be 95% confident that salary increases between about $4,756 and $5,453 for each additional year of experience, holding other features fixed.

PhD (vs. Bachelor's / no Master's) `has_phd`

- The difference between someone with PhD and no PhD nor Master's degree is on average $23,080 (with the other features fixed).
- We can be 95% confident that having a PhD (vs no PhD and no Master's) is associated with a salary increase between about $17,700 and $28,500, holding other features fixed.

Interpretation and Next Steps

Next Steps

- Do more EDA (look at age)
- Collect more data (we only had 373 employees)
- Try other models